



Part A: Conceptual Understanding (Theory)

1. What is Conditional Probability?

Conditional probability is the probability of an event A occurring given that another event B has already occurred. It is denoted as $P(A|B) = P(A \cap B) / P(B)$, provided $P(B) \neq 0$. It quantifies how the occurrence of one event updates the likelihood of another.

2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem states: $P(A|B) = [P(B|A) \times P(A)] / P(B)$. It allows us to update the prior probability of a hypothesis using new evidence. In classification, it is fundamental for computing the posterior probability of a class given the features, enabling probabilistic decision-making (e.g., in Naive Bayes).

3. What assumptions does the Naive Bayes Classifier make?

Naive Bayes assumes that all features are conditionally independent given the class label (the “naive” assumption). It also assumes that each feature contributes equally and independently to the class probability. Despite this strong assumption, it often performs well in practice, especially with high-dimensional data.

4. Explain the working principle of:

K-Nearest Neighbors (KNN):

KNN is a distance-based, non-parametric classifier. For a new sample, it finds the K closest training points (using a distance metric like Euclidean) and assigns the majority class among those neighbors. It is instance-based and makes no assumption about the underlying data distribution.

Support Vector Machine (SVM):

SVM is a margin-based classifier that finds the optimal hyperplane separating classes with the maximum margin. Support vectors are the critical points closest to the boundary. Using kernels, it can handle non-linearly separable data by mapping it to higher dimensions.

5. Compare distance-based, probabilistic, and margin-based classifiers.

Distance-based (e.g., KNN): Classify based on proximity to training instances; simple but computationally expensive at prediction time and sensitive to feature scaling.

Probabilistic (e.g., Naive Bayes): Model the probability distribution of classes and features; provide class probabilities and handle uncertainty well, but rely on distributional assumptions.

Margin-based (e.g., SVM): Optimize the decision boundary for maximum separation; effective in high dimensions and robust to outliers, but less interpretable and can be slower to train.